

INTRODUCTION

As part of the B.Tech curriculum under MG University, we, the Eighth Semester Electrical and Electronics Engineering students of Adi Shankara Institute of Engineering & Technology, Kalady have visited the K.S.E.B 220 kV Substation and Load Dispatch Centre, Kalamassery on 19th January 2016. The IV was successfully completed under the able guidance of Priya miss, Hima miss, Rajesh sir and Divakaran sir.

This training gave us a rare opportunity to understand the power distribution side and to know about the devices employed for that purpose.

Transmission system is an essential link between power stations and load centres for bulk transfer of power and it has a vital role in the management of power. KSEB has pioneered the modern concept of developing a large transmission network for transferring power from power stations to the local load centres. The Kerala power system grid is connected to the Southern Region Transmission system through two 400kV double circuit lines. They are

- (1) Udumalpet- Madakkathara line
- (2) Thirunelveli- Pallippuram (Trivandrum) line.

There are 6 major inter-state transmission lines at 220 KV level and 110 KV level. The 220 KV lines are

- (1) Kaniyampetta- Kadakola(single circuit).
- (2) Idukki – Udumalpet (single circuit).
- (3) Sabarigiri - Theni (single circuit).
- (4) Edamon – Thirunelveli (double circuit).

The 110 KV lines are

- (1).Parassala - Kuzhithura
- (2) Manjeswaram - Konake.

The major substations include two 400 KV sub-stations, and seventeen 220 KV substations. The main grid comprises of the 220 KV systems.

The transmission sector of KSEB comprises of two zones namely North and South, and a System Operation wing each headed by a Chief Engineer. The North zone is headed by the Chief Engineer (Transmission - North) with its head quarters at Kozhikode, and the South zone headed by the Chief Engineer (Transmission-South) with its head quarters at Thiruvananthapuram. The System Operations wing with headquarters at Kalamassery is responsible for the real time management of Kerala Power System and also the activities connected with communication and protection fields.

The State Load Dispatch Centre (SLDC) does meticulous planning and prudent operations to attain excellent results. It includes real time load restrictions as and when necessary, generation scheduling, maintenance scheduling, scheduling the import of power from IPPs and CGS on long term as well as on daily basis, economic load dispatching, merit order dispatching, unit commitment policy, fixing up of merit order for under frequency tripping and remote switching from the sub LD Stations.

The grid is managed by the State Load Dispatch Station situated at Kalamassery. There are 3 Sub Load Dispatch Centres – at Thiruvananthapuram, Kalamassery and Kannur (Kanhirede). The Load Dispatch activities are carried out with the aid of state of art technology computerized Supervisory Control and Data Acquisition System (SCADA). Real-time data from 30 Remote Terminal Units including generating stations and major sub-stations are acquired at the Load Dispatch Station. One more RTU is being installed. The data acquisition is through the communication network installed and maintained by the Board comprising of MW link from Thiruvananthapuram to Kalamassery and FO link from Kalamassery to Kannur. Data from remote stations are collected through Power Line Carrier Communication Network (PLCC) to the nearest nodal station in the broadband network. The state LD Station is connected to the Regional Load Dispatch Centre, Bangalore through FO link for real time data transfer. The real time generation details of all central sector stations are made available through the SRLDC.

The state load dispatch centre schedules generation from various generating stations, central sector stations and IPPs depending on the load condition and the real time frequency. Transaction of unscheduled energy from the pool when the frequency profile is favorable is

coordinated by the SLDC effectively. The load Dispatch Centre also monitors the transmission system and issues sanction for shutdowns. Water availability, inflow, consumption, demand etc. are daily collected and monitored in the Load Dispatch Station. The Load Dispatch activities are co-ordinated by the Chief Engineer, System Operation, Kalamassery. The major activities undertaken by this wing are:

1. Daily scheduling of Generation.
2. Short term and long term planning of generation schedule.
3. Preparation of Load Generation Balance Reports on short term and long term basis.
4. Verification of energy drawals from various central generating stations and Regional Energy Accounts.
5. Verification of energy availed on unscheduled basis from the central grid depending on the frequency.
6. Certification of energy generated from all internal generating stations and the energy received on the grid.
7. Maintenance scheduling of generating units and transmission lines.
8. Economic load dispatching.
9. Grid discipline.
10. Load forecasting and demand estimation.
11. System security and islanding facility.
12. Black start preparedness.
13. Event analysis and preventive measures.
14. Coordination with neighbor Grids.
15. Public relation and consumer interaction.
16. Certification of availability of generating stations, and transmission system.
17. Maintenance of the communication network, communication equipments and SCADA system.
18. Protection coordination, commissioning and troubleshooting of protection schemes at all substations and generating stations.
19. Performance monitoring of major grid elements like power transformers, instrument transformers, generators, capacitor banks, etc.
20. Undertakes testing and commissioning of protection schemes of major EHT consumers on a payment basis.

The Load Dispatch Station activities are technically supported by the communication wing and protection wing under the control of three Dy. Chief Engineers stationed at Thiruvananthapuram, Kalamassery and Kannur.

- Transformer Capacity: 720 MVA (200MVA*3, 120MVA*1)
- Largest Feeder: Idukki, Pallikara
- 110kV is fed to 18 feeders
- 66kV is fed to 8 feeders
- 11kV is fed to 10 feeders
- Radial feeders are used in this substation
- 40 MVAR Capacitor banks are used to achieve unity power factor

COMPONENTS OF SUBSTATION

BUSBARS

Various incoming and outgoing circuits are connected to busbars. Busbars receive power from incoming circuits and deliver power to outgoing circuits.

TRANSFORMERS

The power transformers are used to step down the voltage. The transformer core is made of high grade non-ageing cold rolled grain oriented steel laminations. the cooling is provided by means of ONAN(Oil Natural Air Natural)and ONAF(Oil Natural Air Forced). There are fans rotating at very high speed near to fins (used to increase the surface area). Tap changing mechanism is provided for maintaining the secondary voltage constant with varying primary voltage.

Tap changing causes change in leakage reactance, core loss, I^2R loss etc. the tap changing means changing the turns ratio. The tapings are provided on the HV side so that it carries minimum current.

Tank is of welded construction. It is stiffened by structured steel. It is provided with 2F earth terminals. HV and LV bushings are transported in crates. Conservator is a cylindrical tank divided into 2 chambers –one for the main tank and other for OLTC. The

function of the main conservator is to keep the main tank completely filled with oil. OLTC is fitted with an oil sight window. Explosion vent has a bursting plate and is used as a safety device. When the internal pressure of transformer increases by abnormal phenomenon the diaphragm bursts. In latest models this is replaced by a pressure release valve. Breather: When transformer becomes warm, the oil and gas expands. The gases at the top of the oil expel out and air is drawn into the transformer. This process is called breathing. The air entering is made moisture free by letting it to pass through the breather. It consists of a small container with silica gel crystal. The material is blue when dry and whitish when damp. At the bottom of the container some oil is present along with aluminum silicate in order to absorb dust particles. There are 2 breathers, one for the main conservator tank and the other for OLTC. Two gauges keep track of the temperature of the oil and winding and an alarm goes off when the temperature rises beyond the limited value. The gauges are fixed on the mounting plate of the transformer.

The 2 types of indicators are –Oil Temperature Indicators (OTI)-Winding Temperature Indicators (WTI). There is provision for the control of fan motor by sensing the oil temperature.

LIGHTNING ARRESTER

A lightning arrester is a device used on electrical power systems and telecommunications systems to protect the insulation and conductors of the system from the damaging effects of lightning. The typical lightning arrester has a high-voltage terminal and a ground terminal. When a lightning surge (or switching surge, which is very similar) travels along the power line to the arrester, the current from the surge is diverted through the arrester, in most cases to earth. Smaller versions of lightning arresters, also called surge protectors, are devices that are connected between each electrical conductor in power and communications systems and the Earth.

These prevent the flow of the normal power or signal currents to ground, but provide a path over which high-voltage lightning current flows, bypassing the connected equipment. Their purpose is to limit the rise in voltage when a communications or power line is struck by lightning or is near to a lightning strike.

ISOLATORS

In order to disconnect a part of system for maintenance and repair isolators are used. It is a knife switch designed to open a circuit under no load. If isolators are to be opened, the CB connected must be opened first. Otherwise there is a possibility of occurrence of a spark at the isolator contacts. After repair, first isolators are closed and then CB. There are two types of isolators- line isolators and bus isolators. For bus isolators, there is no earth switch. During maintenance works the line isolator contacts are opened, so that the three phases trip simultaneously. For the ease of earthing, dead weights are provided at the end of earthing arms.

EARTH SWITCH

Earth Switch is used to discharge the voltage on the circuit to the earth for safety. Earth switch is mounted on the frame of the isolators. Earth Switch is located for each incomer transmission line and each side of the busbar section.

CURRENT TRANSFORMER

Current transformers are used for Stepping down current for measurement, protection and control. Current transformers are of two types

1. Protective CT
2. Measuring CT

VOLTAGE TRANSFORMER

Voltage transformers are used to step down the voltage for measurement, protection and control. Voltage transformers are of two types.

1. Electro magnetic type
2. Capacitive VT located on the feeder side of the Circuit Breaker.

WAVE TRAP

Power plants and substations are connected by high voltage power transmission lines which transmit power typically at 50 or 60 Hz. These lines are also used to carry communication and control signals for the operation of the grid (local equipment status, readings, alarms and switching signals). These signals which are of much lower voltage than

the power, use high frequency carriers and wave trap filters are employed to separate the power and communication signals at every receiving end.

CIRCUIT BREAKERS

A circuit breaker is an equipment which can open or close a circuit under normal as well as fault conditions. It can be operated manually from the yard as well as from control room using a pulse trigger under normal condition and automatically under fault condition. For latter relay circuits are used. It consists of a moving and fixed contacts enclosed in strong metal tank. Under normal conditions contacts remain closed and the CT senses full load current. When fault occurs, the over-current in the CT primary winding increases the secondary emf thus energizing the trip coil and moving contacts are pulled down, thus opening the contacts and hence the circuit. The arc produced is quenched by SF₆ (outdoor CB) and vacuum (indoor CB).